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Integrated circuit package and method

Abstract

A method includes attaching semiconductor devices to an interposer structure, attaching the interposer structure to a first carrier substrate, attaching integrated passive devices to the first carrier substrate, forming an encapsulant over the semiconductor devices and the integrated passive devices, debonding the first carrier substrate, attaching the encapsulant and the semiconductor devices to a second carrier substrate, forming a first redistribution structure on the encapsulant, the interposer structure, and the integrated passive devices, wherein the first redistribution structure contacts the interposer structure and the integrated passive devices, and forming external connectors on the first redistribution structure.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE (1) This application is a divisional of U.S. patent application Ser. No. 16/745,610, entitled “Integrated Circuit Package and Method,” filed on Jan. 17, 2020, which application is incorporated herein by reference.

BACKGROUND

(1) Since the development of the integrated circuit (IC), the semiconductor industry has experienced continued rapid growth due to continuous improvements in the integration density of

various electronic components (i.e., transistors, diodes, resistors, capacitors, etc.). For the most part, these improvements in integration density have come from repeated reductions in minimum feature size, which allows more components to be integrated into a given area.

(2) These integration improvements are essentially two-dimensional (2D) in nature, in that the area occupied by the integrated components is essentially on the surface of the semiconductor wafer. The increased density and corresponding decrease in area of the integrated circuit has generally surpassed the ability to bond an integrated circuit chip directly onto a substrate. Interposers have been used to redistribute ball contact areas from that of the chip to a larger area of the interposer. Further, interposers have allowed for a three-dimensional package that includes multiple chips. Other packages have also been developed to incorporate three-dimensional aspects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. 1-2 are cross-sectional views of intermediate steps during a process for forming device packages, in accordance with some embodiments.

(3) FIGS. 3A-C are cross-sectional views and plan views of intermediate steps during a process for forming device packages, in accordance with some embodiments.

(4) FIGS. 4-9 are cross-sectional views of intermediate steps during a process for forming device packages, in accordance with some embodiments.

(5) FIGS. 10-13 are cross-sectional views of intermediate steps during a process for forming device structures, in accordance with some embodiments.

(6) FIG. 14 is a plan view of an intermediate step during a process for forming device packages, in accordance with some embodiments.

(7) FIG. 15 is a schematic diagram of a device package, in accordance with some embodiments.

(8) FIGS. 16A-C are cross-sectional views of intermediate steps during a process for forming package structures, in accordance with some embodiments.

(9) FIGS. 17-22 are cross-sectional views of intermediate steps during a process for forming package structures, in accordance with some embodiments.

(10) FIGS. 23A-C are cross-sectional views of intermediate steps during a process for forming package structures, in accordance with some embodiments.

(11) FIGS. 24-28 are cross-sectional views of intermediate steps during a process for forming package structures, in accordance with some embodiments.

(12) FIG. 29 is a cross-sectional view of a package structure, in accordance with some embodiments.

DETAILED DESCRIPTION

(13) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of

simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(14) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(15) In accordance with some embodiments, packages containing integrated circuit devices are described. In some embodiments, a device structure containing multiple device packages that are electrically connected by a redistribution structure is described. The use of the redistribution structure may allow for smaller features to be formed within each device package, which can improve performance and lithographic patterning. In accordance with some embodiments, a package structure may be formed that includes electronic devices (e.g., surface-mount devices (SMDs)), device packages, and/or device structures. By using two separate carriers to form the package structure, electronic devices having different thicknesses may be used. Additionally a redistribution structure may be used to form electrical interconnections, which can reduce processing cost and the number of processing steps.

(16) FIG. 1 is a cross-sectional view of an integrated circuit device **50**, in accordance with some embodiments. The integrated circuit device **50** may be a logic die (e.g., central processing unit (CPU), graphics processing unit (GPU), system-on-a-chip (SoC), microcontroller, etc.), a memory die (e.g., dynamic random access memory (DRAM) die, static random access memory (SRAM) die, etc.), a power management die (e.g., power management integrated circuit (PMIC) die), a radio frequency (RF) die, a sensor die, a micro-electro-mechanical-system (MEMS) die, a signal processing die (e.g., digital signal processing (DSP) die), a front-end die (e.g., analog front-end (AFE) dies), the like, or combinations thereof. The integrated circuit device **50** may be formed in a wafer, which may include different device regions that are singulated in subsequent steps to form a plurality of integrated circuit devices **50**. The integrated circuit device **50** includes a substrate **52** and an interconnect structure **54**.

(17) The substrate **52** may include a bulk semiconductor substrate, semiconductor-on-insulator (SOI) substrate, multi-layered semiconductor substrate, or the like. The semiconductor material of the substrate **52** may be silicon, germanium, a compound semiconductor including silicon germanium, silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. Other substrates, such as multi-layered or gradient substrates, may also be used. The substrate **52** may be doped or undoped. Devices, such as transistors, capacitors, resistors, diodes, and the like, may be formed in and/or on an active surface (e.g., the surface facing upward) of the substrate **52**.

(18) An interconnect structure **54** having one or more dielectric layer(s) and respective metallization pattern(s) is formed on the active surface of the substrate **52**. The dielectric layer(s) may be inter-metallization dielectric (IMD) layers. The IMD layers may be formed, for example, of a low-K dielectric material, such as undoped silicate glass (USG), phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorosilicate glass (FSG), SiOxCy, Spin-On-Glass, Spin-On-Polymers, silicon carbon material, compounds thereof, composites thereof, combinations thereof, or the like, by any suitable method known in the art, such as spin coating, chemical vapor deposition (CVD), plasma-enhanced CVD (PECVD), high-density plasma chemical vapor deposition (HDP-CVD), or the like. The metallization pattern(s) in the dielectric layer(s) may route electrical signals between the devices, such as by using vias and/or traces, and may also contain various electrical devices, such as capacitors, resistors, inductors, or the like. The various devices

and metallization patterns may be interconnected to perform one or more functions. The functions may include memory structures, processing structures, sensors, amplifiers, power distribution, input/output circuitry, or the like. Additionally, die connectors such as conductive pillars or contact pads, are formed in and/or on the interconnect structure **54** to provide an external electrical connection to the circuitry and devices. One of ordinary skill in the art will appreciate that the above examples are provided for illustrative purposes. Other circuitry may be used as appropriate for a given application.

(19) In some embodiments, the integrated circuit device **50** is a stacked device that includes multiple substrates **52**. For example, the integrated circuit device **50** may be a memory device such as a hybrid memory cube (HMC) module, a high bandwidth memory (HBM) module, or the like, and may include multiple memory dies. In such embodiments, the integrated circuit device **50** includes multiple substrates **52** interconnected by vias. Each of the substrates **52** may (or may not) have a separate interconnect structure **54**.

(20) FIG. **2** is a cross-sectional view of an interposer **70**, in accordance with some embodiments. Although only one interposer **70** is shown, it should be appreciated that the interposer **70** may be formed in a wafer having multiple device regions, with each device region used to form one interposer **70**. The interposer **70** includes a substrate **72**, through vias **74**, and an interconnect structure **76**.

(21) The substrate **72** may be a bulk semiconductor substrate, SOI substrate, multi-layered semiconductor substrate, or the like. The semiconductor material of the substrate **72** may be silicon, germanium, a compound semiconductor including silicon germanium, silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide, an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP, combinations thereof, or the like. Other substrates, such as multi-layered or gradient substrates, may also be used. The substrate **72** may be doped or undoped. Devices, such as transistors, capacitors, resistors, diodes, and the like, may be formed in and/or on a surface of the substrate **72**. The substrate **72** is, in some embodiments, based on an insulating core such as a fiberglass reinforced resin core. For example, the core material may be fiberglass resin such as FR4, bismaleimide-triazine (BT) resin, other printed circuit board (PCB) materials or films, a combination, or the like. Build up films such as Ajinomoto build-up film (ABF) or other laminates may be used for the substrate **72**.

(22) The through vias **74** are formed to extend from the front surface of the substrate **72** into substrate **72**. The through vias **74** are also sometimes referred to as through-substrate vias or through-silicon vias (TSVs) when the substrate **72** is a silicon substrate. The through vias **74** may be formed by forming recesses in the substrate **72** by, for example, etching, milling, laser techniques, a combination thereof, or the like. A thin dielectric material may be formed in the recesses, such as by using an oxidation technique. A thin barrier layer may be conformally deposited over the front side of the substrate **72** and in the openings, such as by CVD, atomic layer deposition (ALD), physical vapor deposition (PVD), thermal oxidation, combinations thereof, and/or the like. The barrier layer may be formed from a nitride or an oxynitride, such as titanium nitride, titanium oxynitride, tantalum nitride, tantalum oxynitride, tungsten nitride, combinations thereof, or the like. A conductive material may be deposited over the thin barrier layer and in the openings. The conductive material may be formed by an electro-chemical plating process, CVD, ALD, PVD, combinations thereof, or the like. Examples of conductive materials are copper, tungsten, aluminum, silver, gold, combinations thereof, or the like. Excess conductive material and barrier layer may be removed from the front side of the substrate **72** by, for example, a chemical-mechanical polish (CMP). Thus, the through vias **74** may include a conductive material, with a thin barrier layer between the conductive material and the substrate **72**.

(23) The interconnect structure **76** is formed over the front surface of the substrate **72**, and is used to form electrical connections between the devices of the substrate **72** (if any), the through vias **74**,

and/or external devices. The interconnect structure **76** may include one or more dielectric layers and respective metallization patterns in the dielectric layers. The metallization patterns may be Redistribution Layers (RDLs) that include vias and/or traces that form the electrical connections. In some embodiments, the interconnect structure **76** may be a redistribution structure or a fan-out structure.

(24) In some embodiments, the interconnect structure **76** is formed by forming a first dielectric layer (not individually labeled in FIG. **2**) on the substrate **72**. In some embodiments, the first dielectric layer is formed of a polymer, which may be a photo-sensitive material such as polybenzoxazole (PBO), polyimide, benzocyclobutene (BCB), or the like, that may be patterned using lithography. In other embodiments, the first dielectric layer is formed of a nitride such as silicon nitride, an oxide such as silicon oxide, Phosphosilicate Glass (PSG), Borosilicate Glass (BSG), Boron-doped Phosphosilicate Glass (BPSG), or the like. The first dielectric layer may be formed by spin coating, lamination, CVD, the like, or a combination thereof. The first dielectric layer is then patterned to form openings that expose the through vias **74**. In embodiments in which the first dielectric layer is formed of a photo-sensitive material, the patterning may be performed by exposing the first dielectric layer in accordance with a desired pattern and performing a developing process to remove the unwanted material, thereby exposing the through vias **74**. Other techniques, such as using a patterned mask and etching, may also be used to pattern the first dielectric layer.

(25) A seed layer (not shown in FIG. **2**) is formed over the first dielectric layer and in the openings formed in the first dielectric layer. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. In some embodiments, the seed layer comprises a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using, for example, PVD or the like. A mask (not shown in FIG. **2**) is then formed and patterned on the seed layer in accordance with a desired metallization pattern. In some embodiments, the mask is a photoresist formed by spin coating or the like, which is then exposed to light for patterning. The patterning forms openings through the mask to expose the seed layer. A conductive material is formed in the openings of the mask and on the exposed portions of the seed layer. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may comprise a metal, such as copper, titanium, tungsten, aluminum, a combination of these, or the like. Then, the photoresist and portions of the seed layer on which the conductive material is not formed are removed. The photoresist may be removed by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. Once the photoresist is removed, exposed portions of the seed layer are removed, such as by using an acceptable etching process, such as by wet or dry etching. The remaining portions of the seed layer and conductive material form a first RDL within the interconnect structure **76**.

(26) Additional dielectric layers and additional RDLs may then be formed over the first dielectric layer to form additional electrical connections within the interconnect structure **76**. The additional dielectric layers and additional RDLs may be formed using similar materials and processes as used to form the first dielectric layer and first RDL. For example, additional dielectric layers may be formed in a process and with materials similar to the first dielectric layer. Openings may be made through each of the additional dielectric layers to expose at least a portion of an underlying RDL. The openings may be formed using a suitable photolithographic mask and etching process, such as those described above for the first dielectric layer, although any suitable process may alternatively be used. In some embodiments, the additional dielectric layers are formed of a photosensitive polymer, and openings may be patterned directly in the additional dielectric layers using a photolithographic mask and etching process.

(27) The additional RDLs may be formed in each additional dielectric layer to provide additional electrical connection within the interconnect structure **76**. In an embodiment, the additional RDLs may be formed using materials and processes similar to the first RDL. For example, a seed layer

may be formed, and a photoresist placed and patterned on top of the seed layer in a desired pattern for an additional RDL. Conductive material may then be formed in the patterned openings of the photoresist using e.g., a plating process. The photoresist may then be removed and the seed layer etched, forming the additional RDL. The interconnect structure **76** may be formed in this manner from multiple dielectric layers and multiple RDLs.

(28) In some embodiments, under bump metallization (UBMs) may be formed and patterned over an uppermost RDL of the interconnect structures **76**. The UB Ms provide electrical connections to the interconnect structures **76** upon which an electrical connector, e.g., a solder ball/bump, a conductive pillar, or the like, may be placed. In some embodiments, the UBMs include a diffusion barrier layer, a seed layer, or a combination thereof. The diffusion barrier layer may include Ti, TiN, Ta, TaN, or combinations thereof. The seed layer may include copper or copper alloys. However, other metals, such as nickel, palladium, silver, gold, aluminum, combinations thereof, and multi-layers thereof, may also be included. The UBMs may be formed using sputtering, electroplating, or the like.

(29) FIGS. **3A** through **15** are cross-section views of intermediate steps during a process for forming stacked semiconductor devices, in accordance with some embodiments. In FIGS. **3A** through **9**, a device package **100** is formed by bonding various integrated circuit devices **50** to the front side of the interposer **70**. In some embodiments, the first device package **100** is a chip-on-wafer (CoW) package, although it should be appreciated that embodiments may be applied to other three-dimensional integrated circuit (3DIC) packages. A device structure **100** may be used in the formation of a package structure such as package structures **200** (see FIG. **22**) or package structure **300** (see FIG. **28**).

(30) In FIGS. **3A-C**, one or more integrated circuit devices **50** are attached to the interposer **70**. FIG. **3A** shows a cross-sectional view of integrated circuit devices **50A** and **50B** attached to an interposer **70**, and FIGS. **3B-C** show plan views of integrated circuit devices **50A** and **50B** attached to the interposer **70** in different arrangements, in accordance with some embodiments. The interconnect structures **54** and **76** physically and electrically connect the integrated circuit devices **50** and the interposer **70**. The integrated circuit devices **50** may be electrically connected to each other through the interconnect structure **76**. The integrated circuit devices **50** may include similar devices and/or different devices. For example, the embodiments shown in FIGS. **3A-C** include integrated circuit devices **50A** and integrated circuit devices **50B**, in which the devices **50A** may have different functions than the devices **50B**. The integrated circuit devices **50A** or **50B** may each have a single function (e.g., a logic device, a memory die, etc.), or may have multiple functions (e.g., a system-on-chip or the like). In an embodiment, the integrated circuit devices **50A** are logic devices such as CPUs and the integrated circuit devices **50B** are memory devices such as HBM modules. In some embodiments, an integrated circuit device **50** may be associated with other integrated circuit devices **50**. For example, a single device **50A** may have one or more device **50B** associated with it and which are electrically connected to that device **50A** (e.g., through the interconnect structure **76**).

(31) The integrated circuit devices **50A** and **50B** may be attached to the interconnect structure **76** using, for example, a pick-and-place tool. In embodiments in which the interposer **70** is formed in a wafer, the integrated circuit devices **50** may be attached in different device regions of the wafer. The different device regions may then be singulated in subsequent steps to form multiple first device packages **100** (see FIG. **9**). The integrated circuit devices **50** may be arranged on the interconnect structure **76** or within a device region in any suitable configuration. For example, FIG. **3B** shows devices **50A** with devices **50B** adjacent one side of each device **50A**, and FIG. **3C** shows devices **50A** with devices **50B** adjacent opposite sides of each device **50A**. In some embodiments, such as the embodiment of FIG. **3B**, the integrated circuit devices **50B** are asymmetrically laid out adjacent the integrated circuit devices **50A**. In some embodiments, such as the embodiment of FIG. **3C**, the integrated circuit devices **50B** are symmetrically laid out adjacent the integrated circuit

devices **50A**. An asymmetric layout may allow the integrated circuit devices **50B** to be located closer to input/output (I/O) connecting regions of the integrated circuit devices **50A**. These are examples, and other configurations or arrangements are possible.

(32) In the embodiment shown in FIG. **3A**, the integrated circuit devices **50A** and **50B** are attached to the interconnect structure **76** with connections that include conductive bumps **102**, conductive bumps **104**, and conductive connectors **106**. The conductive bumps **102** are electrically and physically connected to the interconnect structure **54**, and the conductive bumps **104** are electrically and physically connected to the interconnect structure **76**. The conductive connectors **106** bond the conductive bumps **102** and **104**. The conductive bumps **102** may be formed over UBMs (if present) of the interconnect structure **54** or the interconnect structure **76**. The conductive bumps **102** or **104** may be formed from a conductive material such as copper, aluminum, gold, nickel, palladium, the like, or combinations thereof. The conductive bumps **102** or **104** may be formed by a suitable process such as sputtering, printing, electro plating, electroless plating, CVD, or the like. In some embodiments, the conductive bumps **102** or **104** may also comprise metal pillars (such as copper pillars) formed by sputtering, printing, electroplating, electroless plating, CVD, or the like. The metal pillars may be solder free and have substantially vertical sidewalls or tapered sidewalls. The conductive bumps may also be referred to as microbumps.

(33) The conductive connectors **106** may be formed from a conductive material such as solder, and may be formed by initially forming a layer of solder on the conductive bumps **102** or **104** through methods such as evaporation, electroplating, printing, solder transfer, ball placement, or the like. Once the layer of solder has been formed, a reflow process may be performed in order to shape the conductive connectors **106** into desired bump shapes.

(34) In other embodiments, the integrated circuit devices **50** are attached to the interconnect structure **76** by face-to-face bonds. For example, hybrid bonding, fusion bonding, direct bonding, dielectric bonding, metal bonding, or the like may be used to attach the interconnect structures **54** and **76** without the use of solder. Further, a mix of bonding techniques could be used, e.g., some of the integrated circuit devices **50A** and **50B** could be bonded to the interconnect structure **76** by conductive connectors **106**, and other integrated circuit devices **50A** and **50B** could be bonded to the interconnect structure **76** by face-to-face bonds.

(35) In FIG. **4**, an underfill material **108** is dispensed between the integrated circuit devices **50** and the interconnect structure **76**. The underfill material **108** surrounds the conductive bumps **102** and **104** and the conductive connectors **106**. The underfill material **108** may be any acceptable material, such as a polymer, epoxy, molding underfill, or the like. The underfill material **108** may be formed by a capillary flow process.

(36) In FIG. **5**, an encapsulant **112** is formed on the various components of the structure. The encapsulant **112** may be a molding compound, epoxy, or the like, and may be applied by compression molding, transfer molding, or the like. The encapsulant **112** may be formed over the interconnect structure **76** such that the integrated circuit devices **50** and the underfill material **108** are surrounded and covered by the encapsulant **112**. The encapsulant **112** may then be cured. In some embodiments, excess material of the encapsulant **112** is removed (e.g., by CMP), which may also planarize the structure such that top surfaces of the encapsulant **112** and top surfaces of the integrated circuit devices **50** are level.

(37) In FIG. **6**, the structure is flipped over and attached to a carrier **114** or other suitable support structure for subsequent processing. The carrier **114** may be a glass carrier substrate, a ceramic carrier substrate, or the like. The carrier **114** may be a wafer, such that multiple packages can be formed on the carrier **114** simultaneously. The structure may be attached to the carrier **114**, for example, by a release layer **116**. The release layer **116** may be formed of a polymer-based material, which may be removed along with the carrier **114** from the overlying structures that will be formed in subsequent steps. In some embodiments, the release layer **116** is an epoxy-based thermal-release material, which loses its adhesive property when heated, such as a light-to-heat-conversion (LTHC)

release coating. In other embodiments, the release layer **116** may be an ultra-violet (UV) glue, which loses its adhesive property when exposed to UV lights. The release layer **116** may be dispensed as a liquid and cured, may be a laminate film laminated onto the carrier **114**, or may be the like. The top surface of the release layer **116** may be leveled and may have a high degree of coplanarity.

(38) In FIG. 7, the substrate **72** is thinned (e.g., by CMP) to expose the through vias **74**. In some embodiments, exposed surfaces of the substrate **72** and exposed surfaces of the through vias **74** are level. In some embodiments (not shown), a recessing process may be performed to recess the substrate **72** such that the through vias **74** protrude from the back side of the substrate **72**. The recessing process may be, e.g., a suitable etch-back process using a wet etch and/or a dry etch. In some embodiments, an insulating layer (not shown) may be formed on the back side of the substrate **72**, surrounding and protecting the protruding portions of the through vias **74**.

(39) In FIG. 8, the structure is debonded from the carrier **114** and singulated, forming one or more device packages **100**. In accordance with some embodiments, the debonding includes projecting a light such as a laser light or an ultraviolet (UV) light on the release layer **116** so that the release layer **116** decomposes under the heat of the light and the carrier **114** can be removed. Other techniques of removing the release layer **116** or the carrier **114** may be used in other embodiments. The structure may then be placed on a tape **111** or the like, and then singulated to form individual device packages **100**, as shown in FIG. 8. For example, the interposer **70** may be singulated along scribe line regions between adjacent device regions to form the device packages **100**. The singulation process may include sawing, dicing, or the like. After singulation, the sidewalls of the interposer **70** and the sidewalls of the encapsulant **112** of each device package **100** may be coplanar. An individual device package **100** is shown in FIG. 9. In some embodiments, individual device packages **100** may have a length or a width (e.g., **LW1** shown in FIG. 9) that is between about 20 mm and about 50 mm. The device package **100** may also be referred to as a Chip-on-Wafer (CoW) device.

(40) FIGS. 10 through 16 are cross-section views and plan views of intermediate steps during a process for forming device structures **110** (see FIG. 12), in accordance with some embodiments. Each device structure **110** includes multiple device packages **100** that are electrically connected to each other by a redistribution structure **120**. Similar to the device packages **100** described above, a device structure **110** may be used in the formation of a package structure such as package structure **300** (see FIG. 28). In some embodiments of a device structure **110**, rather than using a single large interposer **70** for a single set of integrated circuit devices **50** (as in device packages **100**), multiple sets of integrated circuit devices **50** are formed on separate, smaller interposers **70** which are electrically interconnected by a redistribution structure **120**. In some cases, forming smaller interposers **70** in this manner allows the use of reticles configured for larger pattern reduction (e.g., reduction of 2×, 3×, 4×, or greater) during photolithographic patterning of the interconnect structure **76**. The use of larger pattern reduction can allow for smaller patterned feature size such as a greater density of RDLs, a smaller linewidth of RDLs, or the like. The larger pattern reduction may also allow for reduced line roughness of patterned features and a reduced chance of process defects during patterning. Additionally, the smaller feature size of the interconnect structure **76** may allow for less noisy and more efficient transmission of electrical signals, particularly for signals at higher frequencies (e.g., greater than about 2 MHz, such as about 5 MHz). Thus, by forming a device structure **110** having multiple integrated circuit devices **50** attached to multiple interposers **70**, electrical interconnects between integrated the semiconductor devices **50** on each interposer **70** may be formed having smaller features sizes and with improved process reliability. In some embodiments, the functionality of a single integrated circuit device (e.g. device **50A** in FIG. 9) may be separated into two or more integrated circuit devices (e.g., devices **50C** and **50D** in FIGS. 15-16), which are each attached to a separate interposer **70**.

(41) Turning to FIG. 10, multiple device packages **100** are attached to a carrier **115**, in accordance

with some embodiments. The carrier **115** may be, for example, a carrier or a material as described previously for the carrier **114** (see FIG. 6). FIG. 10 shows two device packages **100** (designated **100A** and **100B**) attached to the carrier **115**, but in other embodiments, more than two device packages **100** may be attached to the carrier, and the device packages **100** may be attached in any suitable configuration or arrangement. The embodiment shown in FIG. 10 includes a first device package **100A** that includes integrated circuit devices **50A** and **50B** and a second device package **100B** that includes integrated circuit devices **50C** and **50D**. The integrated circuit devices **50** within each device package may be similar or different. For example, the integrated circuit device **50A** in device package **100A** may be similar to or different from the integrated circuit device **50C** in device package **100B**, or the integrated circuit devices **50B** in device package **100A** may be similar to or different from the integrated circuit devices **50D** in device package **100B**. Any combinations of similar or different integrated circuit devices **50** may be present in the device packages **100**, and other configurations of integrated circuit devices **50** are possible.

(42) The device packages **100** may be attached to the interconnect structure **76** using, for example, a pick-and-place tool. In some embodiments, an adhesive layer (not shown in FIG. 10) may be formed on the carrier **115** or on the device packages **100** to facilitate attachment. In some embodiments, individual device packages **100** may have a length or a width (e.g., **LW2** shown in FIG. 10) that is between about 20 mm and about 50 mm. In some embodiments, the device packages **100** may undergo electrical testing before placement on the carrier **115**. In this manner, only known good device packages **100** may be attached, improving the yield of the subsequently formed device structure **110** or package structure **300**.

(43) In FIG. 11, an encapsulant **113** is formed on the device packages **100** and the carrier **115**, in accordance with some embodiments. The encapsulant **113** may be a molding compound, epoxy, or the like, and may be applied by compression molding, transfer molding, or the like. The encapsulant **113** may be similar to the encapsulant **112** (see FIG. 5), in some embodiments. The encapsulant **113** may be formed such that the device packages **100** are surrounded and covered by the encapsulant **113**. The encapsulant **113** may then be cured. In some embodiments, excess material of the encapsulant **113** is removed (e.g., by CMP), which may also planarize the structure such that top surfaces of the encapsulant **113** and top surfaces of the device packages **100** are level.

(44) The redistribution structure **120** is formed over the device packages **100**, and is used to form electrical connections between the device packages **100** and/or external devices. The redistribution structure **120** may include one or more dielectric layers and RDLs that include vias and/or traces that form the electrical connections. The redistribution structure **120** shown in FIG. 12 is an illustrative example, and more or fewer dielectric layers and/or RDLs may be formed in the redistribution structure **120**.

(45) In some embodiments, the redistribution structure **120** is formed in a manner similar to that of the interconnect structure **76** (see FIG. 2). For example, the redistribution structure **120** may be formed by first forming a dielectric layer **122** over the device packages **100** and the encapsulant **113**. In some embodiments, the dielectric layer **122** is formed of a polymer, which may be a photo-sensitive material such as PBO, polyimide, BCB, or the like, that may be patterned using lithographic techniques. The dielectric layer **122** may be formed by spin coating, lamination, CVD, the like, or a combination thereof. The dielectric layer **122** is then patterned to form openings that expose the through vias **74** of the device structures **100**. The patterning may be performed by, for example, exposing the dielectric layer **122** in accordance with a desired pattern and performing a developing process to remove the unwanted material, thereby exposing the through vias **74**. Other techniques, such as using a patterned mask and etching, may also be used to pattern the dielectric layer **122**.

(46) A seed layer (not shown in FIG. 12) is formed over the dielectric layer **122** and in the openings formed in the dielectric layer **122**. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different

materials. In some embodiments, the seed layer comprises a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using, for example, PVD or the like. A mask (not shown in FIG. 12) is then formed and patterned on the seed layer. In some embodiments, the mask is a photoresist formed by spin coating or the like, which is then exposed to light for patterning. The patterning forms openings through the mask to expose the seed layer. A conductive material is then formed in the openings of the mask and on the exposed portions of the seed layer. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may comprise a metal, such as copper, titanium, tungsten, aluminum, a combination of these, or the like. Then, the photoresist and portions of the seed layer on which the conductive material is not formed are removed. The photoresist may be removed by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. Once the photoresist is removed, exposed portions of the seed layer are removed, such as by using an acceptable etching process, such as by wet or dry etching. The remaining portions of the seed layer and conductive material form a RDL 124 within the redistribution structure 120.

(47) Additional dielectric layers and additional RDLs may then be formed over the dielectric layer 122 and RDL 124 to form additional electrical connections within the redistribution structure 120. The additional dielectric layers and additional RDLs may be formed using similar materials and processes as used to form the dielectric layer 122 and/or RDL 124. For example, additional dielectric layers may be formed in a process and with materials similar to the dielectric layer 122. Openings may be made through each of the additional dielectric layers to expose at least a portion of an underlying RDL. The openings may be formed using a suitable photolithographic mask and etching process, such as those described above for the dielectric layer 122, although any suitable process may alternatively be used. In some embodiments, the additional dielectric layers are formed of a photosensitive polymer, and openings may be patterned directly in the additional dielectric layers using a photolithographic mask and etching process.

(48) The additional RDLs may be formed in each additional dielectric layer to provide additional electrical connection within the redistribution structure 120. In an embodiment, the additional RDLs may be formed using materials and processes similar to the RDL 124. For example, a seed layer may be formed, and a photoresist placed and patterned on top of the seed layer in a desired pattern for an additional RDL. Conductive material may then be formed in the patterned openings of the photoresist using e.g., a plating process. The photoresist may then be removed and the seed layer etched, forming the additional RDL. The redistribution structure 120 may be formed in this manner from multiple dielectric layers and multiple RDLs.

(49) Turning to FIG. 13, the carrier 115 is debonded from the structure, forming a device structure 110. In some embodiments, the device structure 110 may also be singulated using e.g., a sawing or dicing process. As a result of the singulation process, the outer sidewalls of the redistribution structure 120 and the outer sidewalls of the encapsulant 113 may be coplanar.

(50) Turning to FIG. 14, a plan view of a device structure 110 is shown, in accordance with some embodiments. The device structure 110 shown in FIG. 14 is representative, and some features are not shown for clarity. The device structure 110 of FIG. 14 shows a particular embodiment using the ideas described with respect to FIG. 13. For example, the device structure 110 includes a device package 100A and a device package 100B, which may be the same as or different than the device packages 100A or 100B described in FIG. 13. The device structure 110 shown in FIG. 14 includes two device packages 100A and 100B, but in other embodiments the device structure 110 may include three or more device packages 100. The device packages 100 may have any suitable configuration or arrangement within the device structure 110.

(51) In some embodiments, within a device structure 110, different device packages 100 may be electrically connected by the redistribution structure 120, and integrated circuit devices 50 within each device package 100 may be electrically connected by the interconnect structure 76 of the device package 100. This is shown in FIG. 14, in which the integrated circuit device 50A of the

device package **100A** is connected to the integrated circuit devices **50B** by the interconnect structure **76** of the device package **100A**, and the integrated circuit device **50C** of the device package **100B** is connected to the integrated circuit devices **50D** by the interconnect structure **76** of the device package **100B**. The connections between integrated circuit devices **50** within each device package **100** are shown in FIG. **14** by a portion **76'** of each interconnect structure **76**. An example portion **76'** is also indicated in FIG. **13**, though the portions **76'** indicated are FIGS. **13-14** is illustrative and may be different than shown. As shown in FIG. **14**, the device package **100A** is electrically connected to the device package **100B** by the redistribution structure **120**. The connections between the device packages **100** are shown in FIG. **14** by a portion **120'** of the redistribution structure **120**. An example portion **120'** is also indicated in FIG. **13**, though the portions **120'** indicated in FIGS. **13-14** are illustrative and may be different than shown.

(52) By using both interconnect structures **76** and a redistribution structure **120** in a device structure **110**, the device structure **110** may incorporate multiple device packages **100** having multiple integrated circuit devices **50**. Due to the use of multiple smaller interposers **76** as described above, the conductive traces (e.g., RDLs) of the interconnect structures **76** may be formed having smaller feature sizes than the conductive traces (e.g., RDLs) of the redistribution structure **120**. For example, features on the interposers **76** may be formed using reticles with larger pattern reduction than the pattern reduction options available for forming features on the redistribution structure **120**. For example, the conductive traces of an interconnect structure **76** may have linewidths between about 0.1 μm and about 3 μm and the conductive traces of a redistribution structure **120** may have linewidths between about 1 μm and about 20 μm . In some cases, the interconnect structures **76** may have better high-frequency performance than the redistribution structure **120** due to the smaller feature sizes of the interconnect structures **76**. Thus, the device structure **110** may allow for improved higher-frequency electrical communication between integrated circuit devices **50** using the interconnect structures **76**, for which higher-frequency operation may be desirable. Additionally, electrical communication between device packages **100** at lower frequencies may be adequate, for which the redistribution structure **120** may be used without negatively impacting performance.

(53) Turning to FIGS. **15-16**, in some embodiments, the functionality of a single integrated circuit device **50** may be separated into multiple integrated circuit devices **50** that are electrically connected by the redistribution layer **120** of the device structure **110**. As an illustrative example, FIG. **15** shows a schematic diagram representing a device package **100'** including a single integrated circuit device **50'** having multiple functional components **51A-D**. The functional components **51A-D** may be, for example, modules, circuits, or the like, or portions thereof or combinations thereof. The functional components **51A-D** shown are illustrative, and an integrated circuit device **50** may have more, fewer, or different types of functional components than described. As shown in FIG. **15**, rather than forming a single larger-area integrated circuit device **50'** on one device package **100'**, the functional components **51A-D** of the integrated circuit device **50'** may be divided between two smaller-area integrated circuit devices **50E** and **50F** on two separate device packages **100E** and **100F**. In this manner, the features of the smaller-area device packages **100E** and **100F** may be formed having smaller size than the features of the larger-area device package **100'**, as described above. Thus, some benefits of smaller feature sizes may be achieved by using multiple integrated circuit devices **50** on multiple device packages **100** within a device structure **110** instead of using a single integrated circuit device **50** on a single device package **100**. In other embodiments, the functionality of a single integrated circuit device **50** may be divided into more than two smaller integrated circuit devices **50**, and, accordingly, the associated device structure **110** may have more than two device packages **100**.

(54) FIGS. **16A** through **22** are cross-section views of intermediate steps during a process for forming a package structure **200** (see FIG. **22**), in accordance with some embodiments. The package structure **200** includes one or more device packages **100** electrically connected to a side of

a redistribution structure **220**, and one or more electronic devices **210** electrically connected to the same side of the redistribution structure **220**. In FIGS. **16A** through **18**, a device package **100** is attached to a first carrier **202** and electronic devices **210** are attached to conductive pads **208** formed on the first carrier **202**. In FIGS. **19** through **20**, the structure is removed from the first carrier **202** and mounted to a second carrier **222**. In FIGS. **21** through **22**, the redistribution structure **220** and external connectors **224** are formed over the device packages **100** and the electronic devices **210**, forming the package structure **200**.

(55) FIGS. **16A-C** show a device package **100**, an electronic device **210**, and a first carrier **202** prior to attachment of the device package **100** and electronic device **210** to the first carrier **202**, in accordance with some embodiments. FIG. **16A** shows a device package **100** with an adhesive **206** formed over the interposer **70** of the device package **100**, in accordance with some embodiments. The device package **100** may be similar to the device packages **100** described previously, such as described in FIG. **9**. The adhesive **206** may be any suitable adhesive, epoxy, die attach film (DAF), or the like.

(56) FIG. **16B** shows an electronic device **210**, in accordance with some embodiments. The electronic device **210** may be, for example, a semiconductor device or other device that includes one or more active devices and/or one or more passive devices such as capacitors, resistors, inductors, and the like. The electronic device **210** may be, for example, an integrated passive device (IPD). In some embodiments, the electronic device **210** is a passive device comprising a capacitor, such as a multi-layer ceramic capacitor (MLCC) or the like. In some embodiments, the electronic device **210** may be a surface-mount device (SMD) or the like. In some embodiments, the electronic device **210** has a thickness between about 50 μm and about 600 μm . The electronic device **210** includes one or more connectors **212** that provide electrical connection between an external component and the electronic device **210**. The connectors **212** may be, for example, conductive bumps, pads, leads, solder balls, or the like.

(57) FIG. **16C** shows a first carrier **202**, which may be a suitable support structure for subsequent processing. The first carrier **202** may be a glass carrier substrate, a ceramic carrier substrate, a wafer, a panel, or the like. The first carrier **202** may be a wafer, such that multiple packages can be formed on the carrier **202** simultaneously. A release layer **204** may be formed on the first carrier **202**. The release layer **204** may be formed of a polymer-based material, which may be removed along with the first carrier **202** from the overlying structures that will be formed in subsequent steps. In some embodiments, the release layer **204** is an epoxy-based thermal-release material, which loses its adhesive property when heated, such as a light-to-heat-conversion (LTHC) release coating. In other embodiments, the release layer **204** may be an ultra-violet (UV) glue, which loses its adhesive property when exposed to UV lights. The release layer **204** may be dispensed as a liquid and cured, may be a laminate film laminated onto the first carrier **202**, a DAF, or the like. The top surface of the release layer **204** may be leveled and may have a high degree of coplanarity.

(58) Still referring to FIG. **16C**, conductive connectors **208** may be formed over the release layer **204**. The conductive connectors **208** may be conductive pads, bumps, pillars, or the like, and are subsequently connected to the connectors **212** of the electronic devices **210** to provide electrical connection to the electronic devices **210**. In some embodiments, the conductive connectors **208** may be formed by depositing a seed layer (not shown) over the release layer **204**. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. In a particular embodiment, the seed layer comprises a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using, for example, PVD or the like. A photoresist is formed and patterned on the seed layer. The photoresist may be formed by spin coating or the like and may be exposed to light for patterning. The pattern of the photoresist corresponds to conductive vias. The patterning forms openings through the photoresist to expose the seed layer. A conductive material is formed in the openings of the photoresist and on the exposed portions of the seed layer. The conductive material

may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may comprise a metal, like copper, titanium, tungsten, aluminum, a combination thereof, or the like. The photoresist and portions of the seed layer on which the conductive material is not formed are removed. The photoresist may be removed by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. Once the photoresist is removed, exposed portions of the seed layer are removed, such as by using an acceptable etching process, such as by wet or dry etching. The remaining portions of the seed layer and conductive material form the conductive connectors **208**.

(59) Turning to FIG. **17**, the device package **100** and the electronic devices **210** are attached to the first carrier **202**, in accordance with some embodiments. FIG. **17** shows a single device package **100** and two electronic devices **210** attached to the first carrier **202**, but more than one device package or more or fewer electronic devices **210** may be attached in other embodiments. The electronic devices **210** attached to the first carrier **202** may be similar electronic devices or may be different electronic devices.

(60) The device package **100** and the electronic devices **210** may be placed on the first carrier **202** using, for example, a pick-and-place tool. The device package **100** may be attached to the release layer **204** on the first carrier **202** by the adhesive **206**. The electronic devices **210** may be attached to the first carrier **202** by the conductive connectors **208**. In some embodiments, the electronic devices **210** may be attached to the conductive connectors **208**, for example, by sequentially dipping the connectors **212** of the electronic devices **210** into a solder material and/or a flux material, and then using a pick-and-place tool in order to physically align the connectors **212** with conductive connectors **208**. In some embodiments, a solder material (e.g., a solder paste) may first be applied to the conductive connectors **208** before placement of the electronic devices **210**. In some cases, after placement of the electronic devices **210**, a reflow may be performed to bond the connectors **212** to the conductive connectors **208**.

(61) In some cases, by attaching the electronic devices **210** to the first carrier **202** in this manner, the electronic devices **210** may be located closer to the device package **100** in the final package structure **200** (see FIG. **22**). For example, in some embodiments, the electronic devices **210** may be separated from the device package **100** by a distance D that is between about $300\text{ }\mu\text{m}$ and about $30,000\text{ }\mu\text{m}$. By locating the electronic devices closer to the device package **100**, electrical resistance may be reduced, and the device performance may be improved, particularly for operation at higher frequencies. Additionally, by attaching the electronic devices **210** to the first carrier, multiple electronic devices **210** having multiple thicknesses may be used in the same package structure **200**, while not increasing the overall thickness of the package structure **200**. The electronic devices **210** may also be protected by an encapsulant **214** (see FIG. **18**), which can improve device reliability.

(62) In FIG. **18**, an encapsulant **214** is formed on the various components of the structure. The encapsulant **214** may be a molding compound, epoxy, or the like, and may be applied by compression molding, transfer molding, or the like. The encapsulant **214** may be formed over the device package **100** and the electronic devices **210** such that the device package **100** and the electronic devices **210** are surrounded and covered by the encapsulant **214**. The encapsulant **214** may then be cured. In some embodiments, excess material of the encapsulant **214** is removed (e.g., by CMP), which may also planarize the top of the structure such that top surfaces of the encapsulant **214** and top surfaces of the device package **100** are level.

(63) In FIG. **19**, a carrier de-bonding is performed to detach (de-bond) the first carrier **202** from the structure, which is then attached to a second carrier **222**. In accordance with some embodiments, the de-bonding includes projecting a light such as a laser light or an ultraviolet (UV) light on the release layer **204** so that the release layer **204** decomposes under the heat of the light and the first carrier **202** can be removed. The structure is then flipped over and attached to a second carrier **222**. The second carrier **222** may be similar to the first carrier **202**, such as comprising a glass carrier

substrate, a ceramic carrier substrate, a wafer, a panel, or the like. A release layer **223** may be formed on the second carrier **222**, and the structure may be attached to the release layer **223**. The release layer **223** may be e.g., a DAF or the like, or may be similar to the release layer **204** described above.

(64) In FIG. **20**, a planarization process is performed on the bottom of the structure, in accordance with some embodiments. The planarization process may be, for example, a CMP. In some embodiments, the planarization process may remove the adhesive **206** and may expose the conductive connectors **208** and the through vias **74** of the device package **100**. In some cases, the planarization process may also planarize the structure such that bottom surfaces of the encapsulant **214** and bottom surfaces of the device package **100** are level.

(65) In FIG. **21**, a redistribution structure **220** is formed over the bottom of the structure, in accordance with some embodiments. The redistribution structure **220** provides electrical interconnection between the electronic devices **210**, the device package **100**, and external components. The redistribution structure **220** includes dielectric layers **226**, **228**, and **230**, and includes metallization patterns **232** and **234**. The metallization patterns may also be referred to as redistribution layers (RDLs) or redistribution lines. The redistribution structure **220** is shown as an example having two layers of metallization patterns. More or fewer dielectric layers and metallization patterns may be formed in the redistribution structure **220**. If fewer dielectric layers and metallization patterns are to be formed, steps and process discussed below may be omitted. If more dielectric layers and metallization patterns are to be formed, steps and processes discussed below may be repeated.

(66) In FIG. **21**, the dielectric layer **226** is deposited on the encapsulant **214**, through vias **74**, and conductive connectors **208**. In some embodiments, the dielectric layer **226** is formed of a photo-sensitive material such as PBO, polyimide, BCB, or the like, which may be patterned using a lithography mask. The dielectric layer **226** may be formed by spin coating, lamination, CVD, the like, or a combination thereof. The dielectric layer **226** is then patterned. The patterning forms openings exposing portions of the through vias **74** and the conductive connectors **208**. The patterning may be by an acceptable process, such as by exposing the dielectric layer **226** to light when the dielectric layer **226** is a photo-sensitive material or by etching using, for example, an anisotropic etch. If the dielectric layer **226** is a photo-sensitive material, the dielectric layer **226** can be developed after the exposure.

(67) The metallization pattern **232** is then formed, in accordance with some embodiments. The metallization pattern **232** includes line portions (also referred to as conductive lines) on and extending along the major surface of the dielectric layer **226**. The metallization pattern **232** further includes via portions (also referred to as conductive vias) extending through the dielectric layer **226** to physically and electrically couple the through vias **74** and the conductive connectors **208**. As an example to form the metallization pattern **232**, a seed layer is formed over the dielectric layer **226** and in the openings extending through the dielectric layer **226**. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. In some embodiments, the seed layer comprises a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using, for example, PVD or the like. A photoresist is then formed and patterned on the seed layer. The photoresist may be formed by spin coating or the like and may be exposed to light for patterning. The pattern of the photoresist corresponds to the metallization pattern **232**. The patterning forms openings through the photoresist to expose the seed layer. A conductive material is then formed in the openings of the photoresist and on the exposed portions of the seed layer. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may comprise a metal, like copper, titanium, tungsten, aluminum, or the like. The combination of the conductive material and underlying portions of the seed layer form the metallization pattern **232**. The photoresist and portions of the seed layer on which the conductive material is not formed are

removed. The photoresist may be removed by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. Once the photoresist is removed, exposed portions of the seed layer are removed, such as by using an acceptable etching process, such as by wet or dry etching. (68) After formation of the metallization pattern **232**, the dielectric layer **228** is deposited on the metallization pattern **232** and dielectric layer **226**. The dielectric layer **228** may be formed and patterned in a manner similar to the dielectric layer **226**, and may be formed of a similar material as the dielectric layer **226**.

(69) The metallization pattern **234** is then formed, in accordance with some embodiments. The metallization pattern **234** includes line portions on and extending along the major surface of the dielectric layer **228**. The metallization pattern **234** further includes via portions extending through the dielectric layer **228** to physically and electrically couple the metallization pattern **232**. The metallization pattern **234** may be formed in a similar manner and of a similar material as the metallization pattern **232**. In some embodiments, the metallization pattern **234** has a different size than the metallization pattern **232**. For example, the conductive lines and/or vias of the metallization pattern **234** may be wider or thicker than the conductive lines and/or vias of the metallization pattern **232**. Further, the metallization pattern **234** may be formed to a greater pitch than the metallization pattern **232**.

(70) The dielectric layer **230** is deposited on the metallization pattern **234** and dielectric layer **228**. The dielectric layer **230** may be formed in a manner similar to the dielectric layer **228**, and may be formed of the same material as the dielectric layer **226**. The dielectric layer **230** is the topmost dielectric layer of the redistribution structure **220**. As such, all of the metallization patterns of the redistribution structure **220** (e.g., the metallization patterns **232** and **234**) are disposed between the dielectric layer **230** and the device package **100**. Further, all of the intermediate dielectric layers of the redistribution structure **220** (e.g., the dielectric layers **226** and **228**) are disposed between the dielectric layer **230** and the device package **100**.

(71) In some embodiments, UBMs **236** are formed for external connection to the redistribution structure **220**. The UBMs **236** may include conductive portions on and extending along the major surface of the dielectric layer **230**. The UBMs **236** further include conductive vias extending through the dielectric layer **230** to be physically and electrically connected to the metallization pattern **234**. The UBMs **236** may be formed of the same material as the metallization pattern **232** or a different material. For example, the UBMs **236** may include copper or copper alloys. However, other metals, such as titanium, nickel, palladium, silver, gold, aluminum, combinations thereof, and multi-layers thereof, may also be included. The UBMs **236** may be formed using sputtering, electroplating, or the like. In some embodiments, the UBMs **236** have a different size than the metallization patterns **232** or **234**.

(72) In some cases, by forming a redistribution structure **220** as described, multiple electronic devices **210** may be connected to at least one device package **100** without the use of a separate interposer or substrate. Additionally, the use of a redistribution structure **220** allows for a package to include components (e.g., device packages **100** or electronic devices **210**) without additional bonding steps (e.g., friction bonding a device package **100** to an interposer), deposition of underfill, or other related processing steps. In this manner, the cost and number of process steps for forming a package may be reduced. Additionally, the use of two carriers (e.g., **202** and **222**) during processing can reduce the amount of warping of the structure during processing.

(73) In FIG. **22**, external connectors **224** are formed on the UBMs **138**. The external connectors **224** may be ball grid array (BGA) connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The external connectors **224** may include a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the external connectors **224** are formed by initially forming a layer of solder through evaporation, electroplating, printing, solder transfer, ball

placement, or the like. Once a layer of solder has been formed on the structure, a reflow may be performed in order to shape the material into the desired bump shapes. In another embodiment, the external connectors **224** comprise metal pillars (such as a copper pillar) formed by a sputtering, printing, electro plating, electroless plating, CVD, or the like. The metal pillars may be solder free and have substantially vertical sidewalls. In some embodiments, a metal cap layer is formed on the top of the metal pillars. The metal cap layer may include nickel, tin, tin-lead, gold, silver, palladium, indium, nickel-palladium-gold, nickel-gold, the like, or a combination thereof and may be formed by a plating process.

(74) Still referring to FIG. **22**, a carrier de-bonding is performed to detach (debond) the second carrier **222** from the structure, forming the package structure **200**. In accordance with some embodiments, the debonding includes projecting a light such as a laser light or an ultraviolet (UV) light on the release layer **224** so that the release layer **224** decomposes under the heat of the light and the second carrier **222** can be removed. The structure may then be flipped over and placed on a tape (not shown). The structure may then be singulated to form the package structure **200**. The singulation may be by sawing, dicing, or the like. As a result of the singulation process, edges of the redistribution structure **220** and the encapsulant **214** may be coplanar.

(75) In this manner, a package structure **200** may be formed with electronic devices **210** having different thicknesses without increasing the overall thickness of the package structure **200**. Additionally, by forming the package structure **200** with a redistribution structure **220** as described herein, the package structure **200** may be formed without the use of an additional interposer or substrate, which can reduce manufacturing cost and reduce the number of processing steps.

(76) FIGS. **23A** through **28** are cross-section views of intermediate steps during a process for forming a package structure **300** (see FIG. **28**), in accordance with some embodiments. The package structure **300** is similar to the package structure **200**, except that an adhesive **206** and conductive connectors **208** are not used. In this manner, the number of processing steps for forming package structure **300** may be fewer than those for forming package structure **200**. The package structure **300** includes one or more device structure **110** electrically connected to a side of a redistribution structure **220**, and one or more electronic devices **210** electrically connected to the same side of the redistribution structure **220**. While the package structure **300** described in FIGS. **23A** through **28** is shown using a device structure **110** (see FIG. **14**), in other embodiments a package structure **300** may use device packages **100** (see FIG. **9**) instead of or in addition to device structures **110**. In FIGS. **23A** through **25**, a device structure **110** and electronic devices **210** are attached to a first carrier **202**. In FIG. **26**, the structure is removed from the first carrier **202** and mounted to a second carrier **222**. In FIGS. **27** through **28**, the redistribution structure **220** and external connectors **224** are formed over the device structure **110** and the electronic devices **210**, forming the package structure **300**.

(77) FIGS. **23A-C** show a device structure **110**, an electronic device **210**, and a first carrier **202** prior to attachment of the device structure **110** and electronic device **210** to the first carrier **202**, in accordance with some embodiments. FIG. **23A** shows a device structure **110**, which may be similar to the device structure **110** described previously, such as described in FIG. **14**. FIG. **16B** shows an electronic device **210**, which may be similar to an electronic device **210** described previously. The electronic device **210** includes one or more connectors **212** that provide electrical connection between an external component and the electronic device **210**. FIG. **16C** shows a first carrier **202**, which may be a suitable support structure for subsequent processing. The first carrier **202** may be similar to the first carrier **202** described previously. A release layer **204** may be formed on the first carrier **202**, which may be similar to the release layer **204** described previously.

(78) Turning to FIG. **24**, the device structure **110** and the electronic devices **210** are attached to the first carrier **202**, in accordance with some embodiments. FIG. **24** shows a single device structure **110** and two electronic devices **210** attached to the first carrier **202**, but one or more device structures **110**, one or more device packages **100**, and/or one or more electronic devices **210** may be

attached in other embodiments. The electronic devices **210** attached to the first carrier **202** may be similar electronic devices or may be different electronic devices, and may have different thicknesses, as shown in FIG. **24**. The device structure **110** and the electronic devices **210** may be placed on the first carrier **202** using, for example, a pick-and-place tool.

(79) In some cases, by attaching the electronic devices **210** to the first carrier **202** in this manner, the electronic devices **210** may be located closer to the device structure **110** in the final package structure **300** (see FIG. **28**). By locating the electronic devices closer to the device structure **110**, electrical resistance may be reduced, and the device performance may be improved, particularly for operation at higher frequencies. Additionally, by attaching the electronic devices **210** to the first carrier, electronic devices **210** having different sizes or different heights may be used, while not increasing the overall thickness of the final package structure **300**. The electronic devices **210** may also be protected by an encapsulant **214** (see FIG. **25**), which can improve device reliability.

(80) In FIG. **25**, an encapsulant **214** is formed on the various components of the structure. The encapsulant **214** may be a molding compound, epoxy, or the like, and may be applied by compression molding, transfer molding, or the like. The encapsulant **214** may be formed over the device structure **110** and the electronic devices **210** such that the device structure **110** and the electronic devices **210** are surrounded and covered by the encapsulant **214**. The encapsulant **214** may then be cured. In some embodiments, excess material of the encapsulant **214** is removed (e.g., by CMP), which may also planarize the top of the structure such that top surfaces of the encapsulant **214** and top surfaces of the device structure **110** are level.

(81) In FIG. **26**, a carrier de-bonding is performed to detach (de-bond) the first carrier **202** from the structure, which is then attached to a second carrier **222**. In accordance with some embodiments, the de-bonding includes projecting a light such as a laser light or an ultraviolet (UV) light on the release layer **204** so that the release layer **204** decomposes under the heat of the light and the first carrier **202** can be removed. The structure is then flipped over and attached to a second carrier **222**. The second carrier **222** may be similar to the first carrier **202** or the second carrier **222** described previously. A release layer **224** may be formed on the second carrier **222**, and the structure may be attached to the release layer **224**. The release layer **224** may be e.g., a DAF or the like, and may be similar to the release layer **224** described above.

(82) In FIG. **27**, a redistribution structure **220** is formed over the bottom of the structure, in accordance with some embodiments. The redistribution structure **220** provides electrical interconnection between the electronic devices **210**, the device structure **110**, and external components. The redistribution structure **220** may be similar to the redistribution structure **220** described previously in FIG. **21**, and may be formed in a similar manner. For example, the redistribution structure **220** shown in FIG. **27** includes dielectric layers **226**, **228**, and **230**, and includes metallization patterns **232** and **234**. More or fewer dielectric layers and metallization patterns may be formed in the redistribution structure **220**. In some embodiments, UBMs **236** are formed for external connection to the redistribution structure **220**, which may be similar to the UBMs **236** described previously.

(83) In some cases, by forming a redistribution structure **220** as described, multiple electronic devices **210** may be connected to at least one device structure **110** without the use of a separate interposer or substrate. Additionally, the use of a redistribution structure **220** allows for a package to include components (e.g., device structures **110**, device packages **100**, and/or electronic devices **210**) without additional bonding steps (e.g., bonding a device structure **110** or a device package **100** to an interposer), deposition of underfill, or other related processing steps. In this manner, the cost and number of process steps for forming a package may be reduced. Additionally, the use of two carriers (e.g., **202** and **222**) during processing can reduce the amount of warping of the structure during processing.

(84) In FIG. **28**, external connectors **224** are formed on the UBMs **236**. The external connectors **224** may be similar to the external connectors **224** described previously. Still referring to FIG. **28**, a

carrier de-bonding is performed to detach (debond) the second carrier **222** from the structure, forming the package structure **300**. In accordance with some embodiments, the debonding includes projecting a light such as a laser light or an ultraviolet (UV) light on the release layer **224** so that the release layer **224** decomposes under the heat of the light and the second carrier **222** can be removed. The structure may then be flipped over and placed on a tape (not shown). The structure may then be singulated to form the package structure **300**. The singulation may be by sawing, dicing, or the like. As a result of the singulation process, edges of the redistribution structure **220** and the encapsulant **214** may be coplanar.

(85) In this manner, a package structure **300** may be formed with electronic devices **210** having different thicknesses without increasing the overall thickness of the package structure **300**. Additionally, by forming the package structure **300** with a redistribution structure **220** as described herein, the package structure **300** may be formed without the use of an additional interposer or substrate, which can reduce manufacturing cost and reduce the number of processing steps. Additionally, the package structure **300** may be formed without the inclusion of conductive connectors **208** or adhesive **206**, further reducing processing cost and steps.

(86) Turning to FIG. **29**, a package structure **400** is shown, in accordance with some embodiments. The package structure **400** includes a device structure **110** that is bonded to an IC substrate **402** by connectors **404**. The device structure **110** may be similar to the device structures described previously, such as that described in FIG. **13**. The IC substrate **402** may be a bulk semiconductor substrate, SOI substrate, multi-layered semiconductor substrate, or the like. The semiconductor material of the IC substrate **402** may be silicon, germanium, a compound semiconductor including silicon germanium, silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide, an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP, combinations thereof, or the like. Other substrates, such as multi-layered or gradient substrates, may also be used. The IC substrate **402** may be doped or undoped. Devices, such as transistors, capacitors, resistors, diodes, and the like, may be formed in and/or on a surface of the IC substrate **402**. For example, the IC substrate **402** may be a semiconductor substrate, chip, die, or the like in which integrated circuits (IC) have been formed, and which are electrically connected to the device structure **110** by the connectors **404**. The IC substrate **402** is, in some embodiments, based on an insulating core such as a fiberglass reinforced resin core. For example, the core material may be fiberglass resin such as FR4, BT resin, other PCB materials or films, a combination, or the like. Build-up films such as ABF or other laminates may be used for the IC substrate **402**.

(87) In some embodiments, the connectors **404** are formed on the redistribution structure **120** of the device structure **110**, and then the device structure **110** is placed on the IC substrate **402** such that the connectors **404** make electrical and physical connection to corresponding conductive features on the IC substrate **402**. The connectors **404** may be similar to the connectors **224** described previously, and may be formed in a similar manner. In some cases, after placement of the device structure **110**, a reflow may be performed to bond the connectors **404** to the corresponding conductive features on the IC substrate **402**. After bonding the device structure **110** to the IC substrate **402**, an underfill **406** may be deposited between the device structure **110** and the IC substrate **402**.

(88) In some embodiments, one or more electronic devices (not shown in FIG. **29**) may also be connected to the IC substrate **402**. The electronic devices may be similar to the electronic devices **210** described previously. The electronic devices may be connected to one side or both sides of the IC substrate **402**. For example, the electronic devices may be attached to the IC substrate **402** on the same side as the device structure **110**.

(89) Embodiments may achieve advantages. By forming a device structure with multiple device packages connected by a redistribution structure, the sizes of each device package may be reduced, allowing for the device packages to be formed having finer features. For example, the metallization

within the interposers of each device package may be formed having greater density and smaller linewidth, which can improve performance at higher frequencies. In some cases, the functionality of a single integrated circuit device can be separated into multiple device packages, which, due to the smaller size of each device package, can allow for improved metallization and improved performance as described. In some cases, the integrated circuit devices may be individually electrically tested before forming the device structure, which can improve yield. By forming a package structure using a two-carrier process, electronic components such as SMDs that have different thicknesses can be incorporated within the same package structure. Additionally, a redistribution structure may be used instead of an interposer within the package structure, which can reduce cost and processing steps.

(90) In an embodiment, a method includes forming a device structure, which includes connecting a first die to a first interposer, connecting a second die to a second interposer, and forming a first redistribution structure over the first interposer and the second interposer, wherein the first redistribution structure electrically connects the first interposer to the second interposer, and forming a package structure, which includes attaching a first surface mounted device (SMD) and the device structure to a first carrier, encapsulating the first SMD and the device structure with a first encapsulant to form an encapsulated structure having a top surface and a bottom surface, attaching the top surface of the encapsulated structure to a second carrier, and forming a second redistribution structure over the bottom surface of the encapsulated structure, wherein the second redistribution structure electrically connects the first SMD and the device structure. In an embodiment, forming the package structure further includes attaching a second SMD to the first carrier, wherein the second SMD has a different thickness than the first SMD. In an embodiment, the method further includes planarizing the encapsulated structure, wherein after planarizing the encapsulated structure, a top surface of the first die and a top surface of the first encapsulant are level. In an embodiment, forming a device structure further includes encapsulating the first interposer and the second interposer with a second encapsulant, wherein the first redistribution structure extends over the second encapsulant. In an embodiment, a thickness of the first SMD is less than a thickness of the encapsulated structure. In an embodiment, the method further includes, after connecting the first die to the first interposer, electrically testing the first die before forming a first redistribution structure over the first interposer. In an embodiment, the sidewalls of the second redistribution structure are coplanar with the sidewalls of the encapsulated structure. In an embodiment, the method further includes forming external connectors on the second redistribution structure.

(91) In an embodiment, a method includes attaching semiconductor devices to an interposer structure, attaching the interposer structure to a first carrier substrate, attaching integrated passive devices to the first carrier substrate, forming an encapsulant over the semiconductor devices and the integrated passive devices, debonding the first carrier substrate, attaching the encapsulant and the semiconductor devices to a second carrier substrate, forming a first redistribution structure on the encapsulant, the interposer structure, and the integrated passive devices, wherein the first redistribution structure contacts the interposer structure and the integrated passive devices, and forming external connectors on the first redistribution structure. In an embodiment, the method further includes forming a second redistribution structure on the interposer structure, wherein the second redistribution structure is attached to the first carrier substrate. In an embodiment, the method further includes forming an interconnect structure on the interposer structure, wherein the semiconductor devices are attached to the interconnect structure. In an embodiment, the method further includes forming an encapsulant over the semiconductor devices before attaching the interposer structure to the first carrier substrate. In an embodiment, the method further includes forming conductive connectors on the first carrier substrate, wherein the integrated passive devices are attached to the first carrier substrate. In an embodiment, attaching the interposer structure to a first carrier substrate includes forming an adhesive on the interposer structure and attaching the

interposer structure to the first carrier substrate using the adhesive. In an embodiment, the integrated passive devices are separated from the interposer structure by the encapsulant.

(92) In an embodiment, a package includes a device substrate and a device structure attached to the first side of the device substrate, the device structure including a first interposer, a second interposer, first semiconductor devices attached to the first interposer, second semiconductor devices attached to the second interposer, and a first redistribution structure connected to the first interposer and the second interposer. In an embodiment, the device substrate includes integrated circuits. In an embodiment, the device substrate is a second redistribution structure. In an embodiment, the device structure further includes a surface-mount device (SMD) attached to a first side of the device substrate. In an embodiment, the package further includes an encapsulant material covering the SMD and surrounding the device structure.

(93) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A package comprising: a device substrate; wherein the device substrate comprises integrated circuits; a first surface-mount device (SMD) attached to the first side of the device substrate; a device structure attached to a first side of the device substrate, the device structure comprising: a first interposer; a second interposer; a plurality of first semiconductor devices attached to the first interposer; a plurality of second semiconductor devices attached to the second interposer; a first redistribution structure connected to the first interposer and the second interposer, wherein the first redistribution structure is attached to the first side of the device substrate with solder bumps; and a first encapsulant on the first redistribution structure, the first encapsulant separating the first interposer from the second interposer; and a second encapsulant covering the first SMD and surrounding the device structure.
2. The package of claim 1, wherein the device substrate is a second redistribution structure.
3. The package of claim 1, wherein the first redistribution structure physically contacts the first side of the device substrate.
4. The package of claim 1, wherein a layer of underfill material is between the plurality of first semiconductor devices and the first interposer.
5. The package of claim 1 further comprising a second surface-mount device (SMD) attached to the first side of the device substrate, wherein the first SMD and the second SMD have different heights.
6. The package of claim 1, wherein top surfaces of the device structure and the second encapsulant are level.
7. The package of claim 1, wherein the second encapsulant extends between the first SMD and the device substrate.
8. A device comprising: a first redistribution structure; a second redistribution structure physically and electrically contacting a first side of the first redistribution structure; a plurality of interposers physically and electrically contacting the second redistribution structure; a respective plurality of semiconductor dies connected to each interposer of the plurality of interposers; a first encapsulant over the second redistribution structure, wherein the first encapsulant encapsulates the plurality of interposers; a first integrated passive device physically and electrically contacting the first side of

the first redistribution structure; a second integrated passive device physically and electrically contacting the first side of the first redistribution structure, wherein the first integrated passive device and the second integrated passive device have different heights above the first side of the first redistribution structure; and a second encapsulant over the first side of the first redistribution structure, wherein the second encapsulant encapsulates the first integrated passive device and the first encapsulant.

9. The device of claim 8, wherein each interposer of the plurality of interposers comprises a respective interconnect structure, wherein the respective plurality of semiconductor dies is connected to the respective interconnect structure.

10. The device of claim 8, wherein the semiconductor dies of each plurality of semiconductor dies are respectively surrounded by a molding material.

11. The device of claim 10, wherein top surfaces of the semiconductor dies and a top surface of the molding material are coplanar.

12. The device of claim 9, wherein the first encapsulant and the second redistribution structure have coplanar sidewalls.

13. The device of claim 8, wherein top surfaces of the second encapsulant and at least one semiconductor die are coplanar.

14. The device of claim 8, wherein the first redistribution structure is wider than the second redistribution structure.

15. A device comprising: a plurality of device packages connected to a first side of a first interconnect structure, wherein each device package comprises: an interposer; a plurality of semiconductor devices connected to the interposer; and a first encapsulant respectively surrounding each semiconductor device of the plurality of semiconductor devices; a second encapsulant respectively surrounding each device package of the plurality of device packages; a device substrate connected to a second side of the first interconnect structure; a passive device connected to the device substrate adjacent the first interconnect structure; and a third encapsulant encapsulating the first interconnect structure, the plurality of device packages, and the passive device, wherein the third encapsulant and the device substrate have coplanar sidewalls.

16. The device of claim 15, wherein the device substrate comprises a second interconnect structure.

17. The device of claim 15, wherein top surfaces of the first encapsulant and the second encapsulant are coplanar.

18. The device of claim 15, wherein each interposer directly contacts the first interconnect structure.

19. The device of claim 15, wherein top surfaces of the first encapsulant and the third encapsulant are coplanar.

20. The device of claim 15, wherein the second encapsulant respectively surrounds each interposer of the plurality of device packages.
